

EXPERIMENT 19: PERFORM EDGE DETECTION USING SOBEL MATRIX ALONG XY AXIS

Aim

To detect edges in both X and Y directions using the Sobel operator in Python and OpenCV.

Algorithm

1. Import OpenCV library.
2. Read the input image.
3. Convert the image to grayscale.
4. Apply Sobel operator along X-axis.
5. Apply Sobel operator along Y-axis.
6. Combine both results using addWeighted().
7. Display and save the output image.

Python Code

```
import cv2

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

sobel_xy = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)
```

```
cv2.imwrite("sobel_xy_output.jpg", sobel_xy)
```

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Sobel XY", sobel_xy)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the edges along both X and Y axes were successfully detected using the Sobel operator.